

Directional Derivative and the Gradient Vector – Part II

Trim Ch 11, S. 12.8, 12.4

Outline:

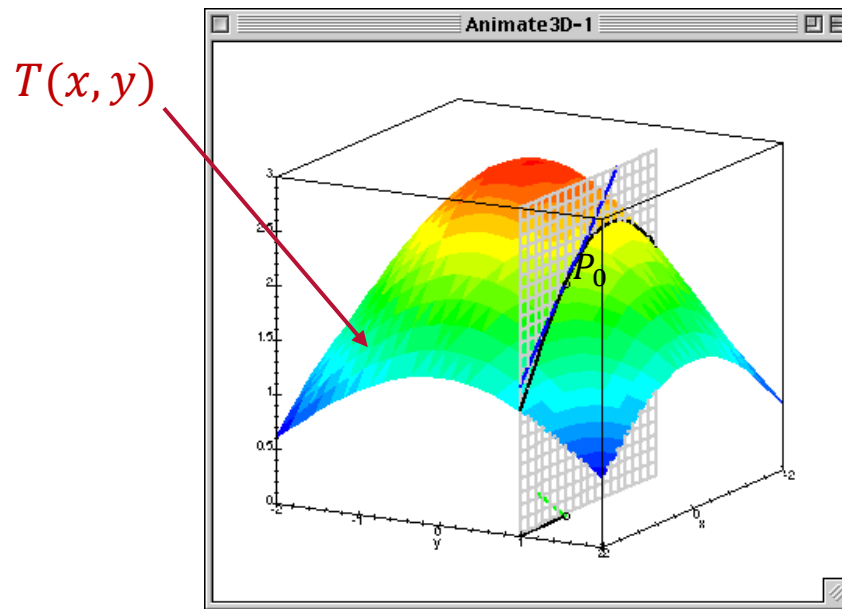
1. Properties of the directional derivative and the gradient vector



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Directional Derivative – Physical Interpretation

If $T(x, y)$ is the temperature at each point $P_0(x_0, y_0)$ of the surface, defined over a region R ,



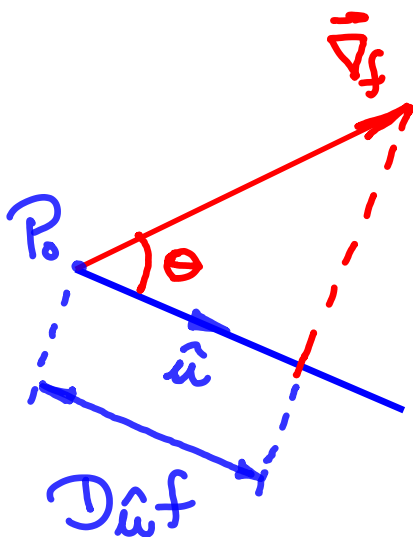
The directional derivative is the **instantaneous rate of change** of temperature at P_0 **stepping off in the direction of $\hat{u}$**

$$D_{\hat{u}}T|_{P_0} = \left. \frac{dT}{ds} \right|_{\hat{u}, P_0} = \vec{\nabla}T|_{P_0} \cdot \hat{u}$$

Directional Derivative in the Direction of $\hat{u}$

The directional derivative of a function $f(x, y)$ at P_0 in the direction of $\hat{u}$ gives us insight into some of the properties of the gradient vector,

$$D_{\hat{u}}f|_{P_0} = \left. \frac{df}{ds} \right|_{\hat{u}, P_0} = \vec{\nabla}f|_{P_0} \cdot \hat{u}$$



$$|D_{\hat{u}}f| = |\vec{\nabla}f| |\hat{u}| \cos\theta$$

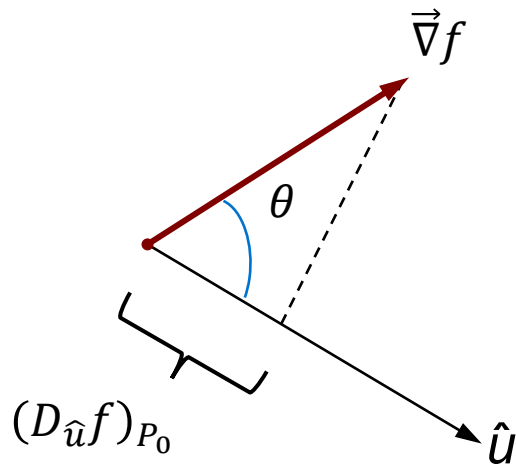
$\underbrace{|\hat{u}|}_{=1}$

$$D_{\hat{u}}f = |\vec{\nabla}f| \cos\theta$$

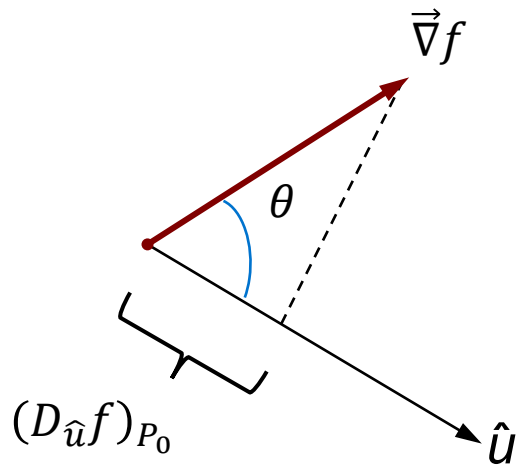
The directional derivative $D_{\hat{u}}f|_{P_0}$ is the component of the gradient vector at P_0 in the direction of $\hat{u}$.



Maximum and Minimum Rate of Change

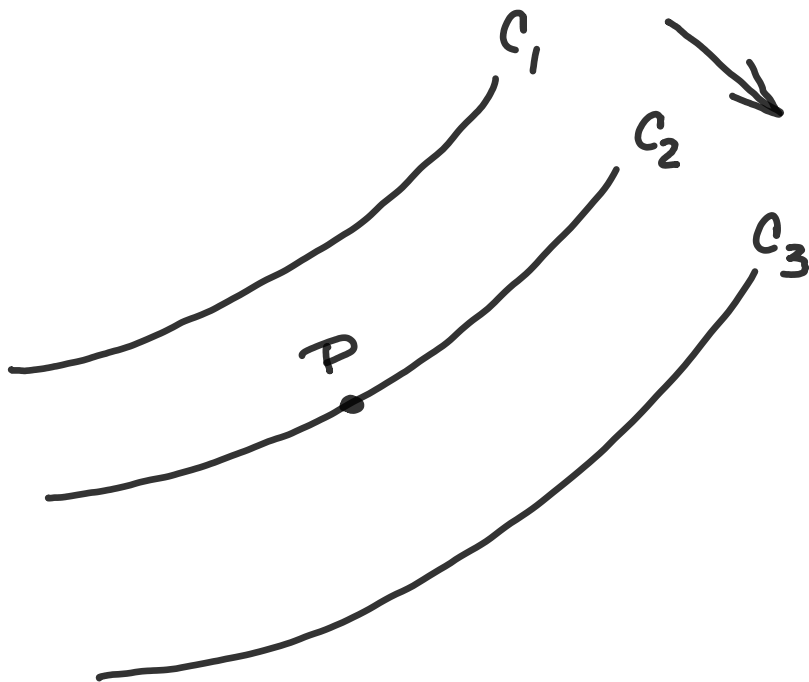


Direction of Zero Change



Direction of the Gradient Vector $(\vec{\nabla} f)_{P_0}$

$\vec{\nabla} f$ can be observed on a contour map:



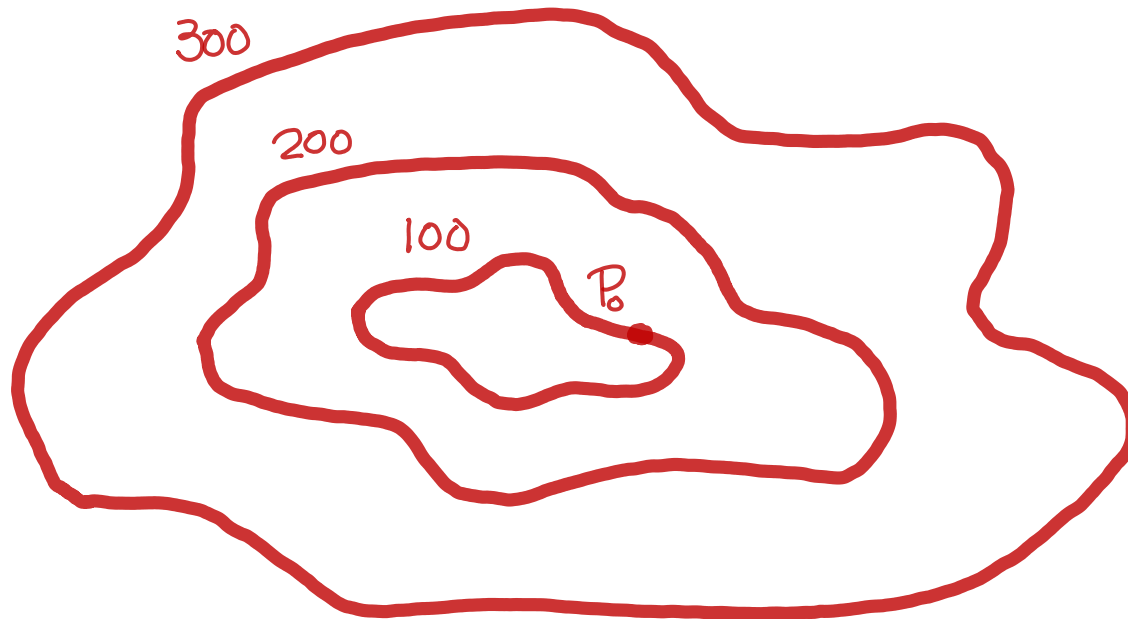
1. When is the rate of change from P to N the largest?
2. When is the distance between PN the smallest?
3. If f increases in most rapidly in the direction of $\vec{\nabla} f$, at P, in which direction will $\vec{\nabla} f$ be pointing?

$\vec{\nabla} f$ is in the direction of the outward normal $\hat{n}$ to the level curve.



Direction of the Gradient Vector $(\vec{\nabla} f)_{P_0}$

How do I travel from an elevation of 100 meters to an elevation of 300 meters the fastest?



Direction of the Gradient Vector $(\vec{\nabla} f)_{P_0}$

$\vec{\nabla} f$ can also be observed on a family of surface levels:

The gradient represents, in magnitude and direction, the greatest rate of increase (decrease) of f with distance

In our example ...

The gradient of the function $f(x, y) = x^3 - 3xy + 4y^2$ was found as,

$$\vec{\nabla} f = (3x^2 - 3y)\hat{i} + (-3x + 8y)\hat{j}$$

- a. What is the direction of maximum increase at the point (1,2)

- b. What is the direction of zero change at (1,2)



Problem

Calculate the directional derivative of the function,

$$f(x, y, z) = x^2y + y^2z + z^2x$$

at the point $(1, -1, 0)$ along the line,

$$x + 2y + 1 = 0$$

$$x - y + 2z = 2$$

in the direction of decreasing z .

Problem

Problem

What is the direction of maximum increase of $f(x, y, z)$ at $(1, -1, 0)$?

What is the direction at which we need to travel for zero change?

Summary

- **The directional derivative is the instantaneous rate of change of a function $f(x, y)$ at P stepping off in the direction of $\hat{u}$**
- The directional derivative is the component of the gradient vector in the direction of $\hat{u}$
- The function $f(x, y)$ increases (decreases) most rapidly in the direction of $\vec{\nabla}f$ ($-\vec{\nabla}f$)
- Any direction $\hat{u}$ perpendicular to $\vec{\nabla}f$ is a direction of zero change
- The gradient represents, in magnitude and direction, the greatest rate of increase/decrease of $f(x, y)$ with distance